

CSCI 332, Fall 2026

Homework 6

Due Monday, October 5 Anywhere on Earth (6am Tuesday)

Submission Requirements

- Type or clearly hand-write your solutions into a PDF format so that they are legible and professional. Submit your PDF on Gradescope.
- Do not submit your first draft. Type or clearly re-write your solutions for your final submission. If your submission is not legible, we will ask you to resubmit.
- Use Gradescope to assign problems to the correct page(s) in your solution. If you do not do this correctly, we will ask you to resubmit.

Academic Integrity

Remember, you may access **any** resource in preparing your solution to the homework. However, you **must**

- write your solutions in your own words, and
- credit every resource you use (for example: “Bob Smith helped me on this problem. He took this course at UM in Fall 2020”; “I found a solution to a problem similar to this one in the lecture notes for a different course, found at this link: www.profzeno.com/agreatclass/lecture10”; “Here is a link to the full transcript of my chat with ChatGPT about the problems on this HW”.) Ask if you need any clarifications about how to properly cite your sources!

1. (0 points) List the resources you used for each problem in this homework. If you used no resources except lecture notes and videos and the linked textbook, you can say “none”. If you used an AI tool, you should put a link to your full conversation here, and you must use the [provided prompt](#) to guide the AI to act as a tutor, not an answer-provider.

You will be asked to resubmit your homework if this section is blank.

Remember that no matter what resources you consulted, you must write your solutions *in your own words*.

2. (3 points)

- (a) Recall the algorithm for binary search of sorted array $A[1..n]$ to find the index of a target element t , or -1 if it does not occur:

```

BinarySearch(sorted  $A[1..n]$ ,  $t$ ):
   $\ell = 1, r = n$ 
  while  $\ell \leq r$ :
     $m = \lfloor \frac{\ell+r}{2} \rfloor$ 
    if  $A[m] > t$ : ⟨go left⟩
       $r = m - 1$ 
    else if  $A[m] < t$ : ⟨go right⟩
       $\ell = m + 1$ 
    else: ⟨equal⟩
      return  $m$ 
  return  $-1$ 

```

In the worst case, how many times does the while loop execute? Give a Θ bound.

- (b) Now consider another problem regarding arrays: suppose you are given an array $A[1..n]$. It is not (necessarily) sorted, but it has another property: it alternates even and odd integers, except at one spot, where two even integers are adjacent. For example, in the array below, 88 and 62 are the adjacent even integers; notice that every other adjacent pair contains one even and one odd.

17	40	23	72	39	88	62	13	40	53	92	21	10	73	68
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Give an $O(\log n)$ time algorithm to find the index i that starts the even pair; that is, the index i so that $A[i] + A[i + 1]$ is even. (Recall that the sum of integers that are both even are even, whereas the sum of integers where one is even and one is odd is odd.) In the example above, $i = 6$.

The array may start either with an even or odd integer, but it satisfies the properties listed above.

3. (7 points) Describe recursive algorithms for the following variants of the text segmentation problem. Assume that you have a subroutine `IsWORD` that takes an array of characters as input and returns `TRUE` if and only if that string is a word.
 - (a) Given an array $A[1..n]$ of characters, compute the number of unique partitions of A into words. For example, given the string **ARTISTOIL**, your algorithm should return 2, for the partitions **ARTIST OIL** and **ART IS TOIL**.

- (b) Given two arrays $A[1..n]$ and $B[1..n]$ of characters, decide whether A and B can be partitioned into words at the same indices. For example, the strings **BOTHEARTHANDSATURNSPIN** and **PINSTARTRAPSANDRAGSLAP** can be partitioned into words at the same indices as follows:

BOT HEART HAND SAT URNS PIN
PIN START RAPS AND RAGS LAP

- (c) Given two arrays $A[1..n]$ and $B[1..n]$ of characters, compute the number of different ways that A and B can be partitioned into words at the same indices.